Declaration of AI assistance

Prepared for Finite Fields and Their Applications

Manuscript: The Welch function is not third-order sum-free

Authors: Deep Bhattacharjee

Research and manuscript preparation. The existing provenance record describes the use of Claude (Anthropic), through Claude Code under Deep Bhattacharjee's direction, for literature searches, development and checking of arguments, verification code, and drafting of the manuscript. This assistance extended beyond spelling and grammar. The human authors retain responsibility for the content.

Preparation of this submission package. OpenAI Codex assisted on 26 October 2026 with source and metadata alignment, compilation, layout review, preparation of submission materials, and release packaging. The record of this work distinguishes fresh checks from results reported by the existing repository.

Verification scope. Lean checks supplied algebraic identities and finite-field certificates for $n = 5$ and $n = 7$. The complete general trace and character-sum proof is written mathematics; it is not a complete Lean formalisation. Finite computations are supporting checks, not the proof for all n. The successful prior repository CI is recorded in verification/previous-ci.json; no new local Lean rebuild is claimed by this package.

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